Jaypee Univeristy of Information Technology, Waknaghat Solan H.P

Tutorial Sheet
Red Black Tree

Q1) State all the rules a red black tree must follow.

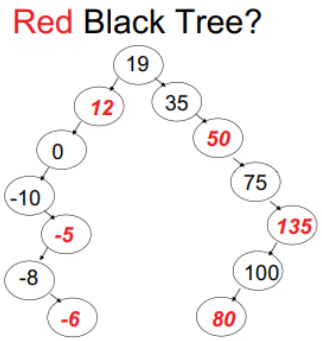
Q2) When it would be optimal to prefer Red-black trees over AVL trees?

a) when there are more insertions or deletions
b) when more search is needed
c) when tree must be balanced
d) when log(nodes) time complexity is needed

Q3) How can you save memory when storing color information in Red-Black tree?

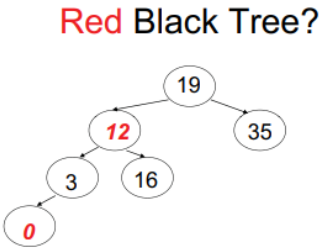
a) using least significant bit of one of the pointers in the node for color information
b) using another array with colors of each node
c) storing color information in the node structure
d) using negative and positive numbering

Q4) Is the tree below a binary search tree? Is it a red black tree?



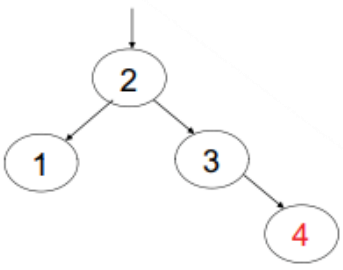
a) BST – Yes, RBT - Yes
b) BST – Yes, RBT - No
c) BST – No, RBT - Yes
d) BST – No, RBT – No

Q5) Is the tree below a binary search tree? Is it a red black tree?

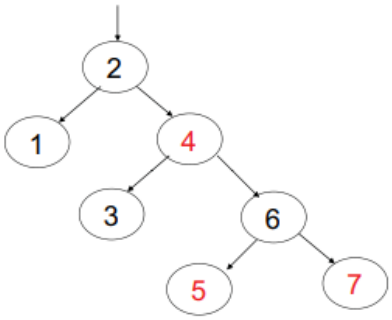


a) BST – Yes, RBT - Yes
b) BST – Yes, RBT - No
c) BST – No, RBT - Yes
d) BST – No, RBT – No

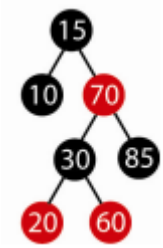
Q6) Insert a node with value 5 in the given RBT



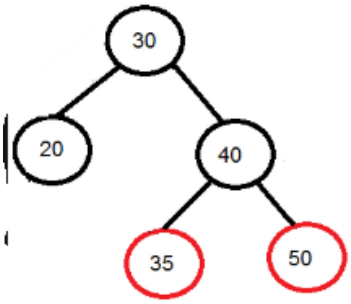
Q7) Insert a node with value 8 in given RBT



Q8) Insert a node with value 50 in the given RBT



Q9) Delete a node with value 20 in the given RBT



Q10) Delete a node with value 20 in the given RBT

